NON-TRIVIAL GENERALIZATIONS OF TUKEY DEPTH FOR DISTRIBUTION FAMILIES AND ANALYSIS OF RANDOMIZED APPROXIMATIONS

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Relevance. Tukey depth is a cornerstone measure of centrality in multivariate analysis. Exact computation is challenging for many distributions of interest, motivating both theoretical generalizations and efficient numerical methods, e.g., for uniform distributions on unit balls and stable laws.

Objective. We extend Tukey depth to two non-trivial classes: (i) uniform (isotropic) distribution on the unit ball $B^p \subset \mathbb{R}^p$, and (ii) the family of symmetric α-stable distributions. Furthermore, we analyze randomized approximation algorithms for depth computation in higher dimensions.

Main Results. In the spherical Gaussian case, we prove that for $\theta \in B^p$,

$$\operatorname{depth}(\theta) = \frac{1}{2}\Big[1 - I_{\|\theta\|^2}\big(\tfrac{1}{2}, \tfrac{p+1}{2}\big)\Big],$$

where $I_z(a, b)$ is the regularized incomplete Beta function. For symmetric α-stable laws in $\mathbb{R}^2$, we show that the optimal directional threshold satisfies

$$c_{\max}(u, v) = \begin{cases} (u^\beta + v^\beta)^{1/\beta}, & \alpha > 1, \\ \max\{u, v\}, & 0 < \alpha \le 1, \end{cases}$$

with $\beta = \alpha/(\alpha - 1)$. With use of numerical experiments we generate Tukey depth contours for the bivariate t- and exponential distributions, illustrating parameter effects on contours' shapes.

Future Work. Building on randomized approximation theory [1], we will conduct comprehensive computational studies to assess approximation errors for intermediate depths, benchmark algorithmic complexity, and refine methods for robust multivariate data analysis.

References

[1] S. Briend, G. Lugosi, and R.I. Oliveira. *On the quality of randomized approximations of Tukey's depth.* arXiv:2309.05657 (2023).

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